Lecture 11. RELATIONS.

SETS → Syntax: $\{\}$.

SETS → Semantic: Collection of Unique elements.

1- Union.
2- Intersection.
3- Subtraction.
4- Complement.
5- Cardinality.
6- Subset.
7- Power Set.
8- Set multiplication.

$A = \{a, b\}$. $B = \{1, 2\}$.

$A \times B = \{\boxed{(a,1)}, (a,2), (b,1), (b,2)\}$.

$B \times A = \{(1,a)(1,b), (2,a), (2,b)\}$. $A \times B \neq B \times A$.

$|A \times B| = |A| \times |B| = 2 \times 2 = \underline{4}$.

$Pow(A) = \{\phi, \{a\}, \{b\}, \{a,b\}\}$.

$|Pow(A)| = 2^{|A|} = 2^2 = 4$.

$A = \{a, b, c\}$.

$Pow(A) = \{\phi, \{a\}, \{b\}, \{c\}, \{a,b\}, \{b,c\}, \{a,c\}, \{a,b,c\}\}$.

$|Pow(A)| = 2^{|A|} = 2^3 = 8$.

$A = \{a, b\}$.
$A = \{a, b\}$.

$A \times A = \{(a,a), (a,b), (b,a), (b,b)\}$.

$|P(A \times A)| = 2^{|A \times A|} = 2^{|A| \times |A|} = 2^{2 \times 2} = 2^4 = 16$

$Pow(A \times A) = \{\phi, \{(a,a)\}, \{(a,b)\}, \{(b,a)\}, \{(b,b)\},$
$\{(a,a), (a,b)\}, \{(a,a), (b,a)\}, \{(a,a), (b,b)\},$
$\{(a,b), (b,a)\}, \{(a,b), (b,b)\}, \{(b,a), (b,b)\},$

$\{(a,a),(a,b),(b,a)\}$, $\{(a,a),(a,b),(b,b)\}$,

$\{(a,a),(b,a),(b,b)\}$, $\{(a,b),(b,a),(b,b)\}$,

$\{(a,a),(a,b),(b,a),(b,b)\}$, $\}$.

Relations.

Binary Relations: R on $A \times B$ is Any Subset of $A \times B$. $R \subseteq A \times B$.

$A = \{a, b\}$, $A \times A = \{(a,a),(a,b),(b,a),(b,b)\}$.

$Pow(A \times A) = \{ \phi, \{(a,a)\}, \{(a,b)\}, \{(b,a)\}, \{(b,b)\},$

$\{(a,a),(a,b)\}, \{(a,a),(b,a)\}, \{(a,a),(b,b)\},$

$\{(a,b),(b,a)\}, \{(a,b),(b,b)\}, \{(b,a),(b,b)\},$

$\{(a,a),(a,b),(b,a)\}, \{(a,a),(a,b),(b,b)\},$

$\{(a,a),(b,a),(b,b)\}, \{(a,b),(b,a),(b,b)\},$

$\{(a,a),(a,b),(b,a),(b,b)\}, \}$.

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$R = \{(a,b) \mid a \le b\}$. $A = \{1,2,3,4\}$.

$= \{(1,1), (1,2), (1,3), (1,4)$
$(2,2), (2,3), (2,4)$
$(3,3), (3,4), (4,4)\}$.

$A \times A = \{(1,1),(1,2),(1,3),(1,4),$
$(2,1),(2,2),(2,3),(2,4),$
$(3,1),(3,2),(3,3),(3,4)$

$(a,2), (a,3), (a,4)$
$(3,3), (3,4), (4,4)\}$,

$(3,1), (3,2), (3,3), (3,4)$
$(4,1), (4,2), (4,3), (4,4)\}$,

$R_2 = \{(a,b) \mid a > b\}$,
$R_3 = \{ \quad " \quad \mid a = b \}$
$R_4 = \{ \quad " \quad \mid a = b+1\}$
$R_5 = \{ \quad " \quad \mid a+b \leq 1\}$.

HW.

Properties of Relations.

1- Reflexive $\forall a \in A \quad \underline{(a,a) \in R}$.

Ex 7 :-
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$R_1 = \{(1,1)\}$ X.

$(1,1) \in R_1 \wedge (2,2) \in R_1 \wedge (3,3) \in R_1 \wedge (4,4) \in R_1$
$T \wedge F \wedge F \wedge F = F$.

$R_2 = \{ \}$. X. $\qquad A = \{1,2,3,4\}$.

$R_3 = \{(1,1), (2,2), (2,1), (3,2), (3,3), (4,4)\}$. ✓

$A = \phi \qquad A \times A = \phi$.
$\mathrm{Pow}(A \times A) = \{\phi\}$. ✓

1- Reflexive $\underline{\forall a \in A \quad (a,a) \in R}$.

$A = \{a\} \qquad A \times A = \{(a,a)\}$.
$\mathrm{Pow}(A \times A) = \{\phi, \{(a,a)\}\}$.
X ✓

$A = \{a,b\}$. $\qquad A \times A = \{(a,a), (a,b), (b,a), (b,b)\}$.

Reflexive :- $\forall a \in A \quad (a,a) \in R.$

$(a,a) \in R \wedge (b,b) \in R.$

$Pow(A \times A) = \{ \phi, \{(a,a)\}, \{(a,b)\}, \{(b,a)\}, \{(b,b)\},$
$\{(a,a),(a,b)\}, \{(a,a),(b,a)\}, \{(a,a),(b,b)\},$
$\{(a,b),(b,a)\}, \{(a,b),(b,b)\}, \{(b,a),(b,b)\},$
$\{(a,a),(a,b),(b,a)\}, \{(a,a),(a,b),(b,b)\},$
$\{(a,a),(b,a),(b,b)\}, \{(a,b),(b,a),(b,b)\},$
$\{(a,a),(a,b),(b,a),(b,b)\}, \}.$

Symmetric : $\forall a,b \in A$ if $(a,b) \in R \rightarrow (b,a) \in R.$

$A = \{1,2,3,4\}.$

Ex 7
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$R_1 = \{ \ \}$ ✓.

$R_2 = \{(1,1)\}$ ✓.

$R_3 = \{(1,1),(1,2)\}$ X.

$R_4 = \{(1,2),(2,1),(3,1)\}$ X.

$A = \phi \qquad A \times A = \phi.$
$Pow(A \times A) = \{\phi\}$ ✓

$A = \phi$ $\quad A \times A = \phi.$

$pow(A \times A) = \{\phi\}.$ ✓

$A = \{a\}$ $\quad A \times A = \{(a,a)\}.$

$pow(A \times A) = \{\phi, \{(a,a)\}\}$

$A = \{a,b\}.$

$pow(A \times A) = \{\phi, \{(a,a)\}, \{(a,b)\}, \{(b,a)\}, \{(b,b)\},$
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$\{(a,a),(a,b),(b,a)\}, \{(a,a),(a,b),(b,b)\},$
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$\{(a,a),(a,b),(b,a),(b,b)\}, \}.$

Quiz #3 $\qquad$ 2-Oct-2025.

$R = \{(a,b) \mid a \div b = Z\}.$ $\qquad A = \{4, 8, 10, 5, 15, 20, 25\}.$

1) find R (5). $\qquad A \times A$ X.

2) Is R Reflexive? (if Yes Say Yes if Not give Counter Example).

3) Is R Symmetric? (if Yes Say Yes

3) Is R Symmetric? (if Yes Say Yes, if Not give Counter Example).

Solution:-

$$A = \{4, 8, 10, 5, 15, 20, 25\}.$$

$$R = \{(4,4), (8,4), (8,8), (10,10), (10,5), (5,5), (20,5), (15,5), (25,5), (20,4), (20,20), (15,15), (25,25), (20,10), (25,5)\},$$

Reflexive. ✓

Symmetric:- $(20,4) \in R \rightarrow (4,20) \notin R$.